Reproduction of Table I of arXiv:1708.02243

S. Knapen, H. K. Lou, M. Papucci, J. Setford, “Tracking down quirks at the Large Hadron Collider”, Phys. Rev. D 96, 115015 (2017) · signal efficiencies, backgrounds and Figures 2–5 · 2000-event samples · 9 October 2026

Summary

- **Trigger and fiducial:** colored trigger 207/873 = 0.24 and fiducial 22/68 = 0.32 agree with the paper (0.24, 0.28). The EW trigger 186/1045 = 0.18 is far above the paper's 0.10; the EW fiducial 62/186 = 0.33 agrees with 0.28.
- **Reconstruction with the SCT z resolution of the paper's text (1.16 mm):** colored 1/22, EW 10/62, far below the paper (0.35, 0.72).
- **Reconstruction without SCT z smearing:** colored 8/22 = 0.36 and EW 50/62 = 0.81, consistent with the paper. Only this setting matches the paper's Fig. 5 Δs values; over the Λ column it fits far better than 1.16 mm, but EW at low Λ still differs.
- **Backgrounds** with the cited detector model [34] and SCT z 580 μm: 0 of 9139 converted photons + pile-up and 0 of 5000 pile-up events in the signal region, as in the paper. If the background is also simulated without SCT z smearing, 146 of 9139 conversions (1.6%) enter the signal region.
- **Figures 2–5** have the paper's shapes; the signal strip width Δw is about 2× (EW) and 5× (colored) wider than in the paper's Fig. 5.

Verdicts use the exact binomial p-value against the paper's number, taken as exact: $p < 0.01$ strongly inconsistent; 0.01–0.05 statistically inconsistent; 0.05–0.10 borderline / mild tension; $p \geq 0.10$ no statistically significant inconsistency. Intervals are Wilson 95%.

1. Simulation setup

1.1 Event generation (both samples generated with the same setup)

Setting	This work	Paper
Generator	MadGraph5_aMC@NLO 3.7.3 + Pythia 8.317	MadGraph5_aMC@NLO + Pythia8 (source comments: v2.3.3, 8.212)
Processes	colored: p p → t1 t1~ (+ j); EW: p p → qe+ qe− (+ j)	vector-like quirks, up to 1 jet; colored QCD only; EW Drell–Yan, unit charge
Masses, Λ	1800 GeV / 2 keV; 800 GeV / 4 keV	same benchmarks
Beams, PDF, scale	pp 13 TeV; nn23lo1 (built-in; LHAPDF 247000); MG5 default dynamical scale	not stated
Matching	MLM, ickkw = 1, xqcut = 30 GeV, qCut = 45 GeV, nJetMax = 1	“matched using Pythia8”; settings not stated
Shower, hadronisation	Pythia 8 defaults (MPI on); colored: R-hadrons on, stable	colored quirks hadronise into quirk-hadrons
Events, seeds	2000 each; MG5 iseed 8001 (colored), 8002 (EW)	not stated

1.2 Cross sections and event statistics

	Colored 1.8 TeV	EW 800 GeV
σ, MadGraph (0 + 1 jet, before matching)	0.753 ± 0.004 fb	0.454 ± 0.002 fb
σ after MLM matching	0.329 ± 0.009 fb	0.237 ± 0.005 fb
Events generated	2000	2000
Events after matching	873	1045
pT(QQ̄) > 200 GeV (trigger)	207	186
both quirk-hadrons charged (colored)	68	186 (unit charge)
pT(QQ̄), all matched events: median [16–84%] (GeV)	77 [19–272]	60 [15–223]
pT(QQ̄), triggered events (GeV)	342 [232–648]	328 [234–575]
M(QQ̄), all matched events (GeV)	4085 [3775–4660]	1934 [1717–2409]

The trigger is the paper's pT(QQ̄) > 200 GeV on the final-state quirk pair (colored: the final-state R-hadrons).

1.3 Detector model and efficiency definitions

Item	This work	Source
Layers	IBL + 3 pixel + 4 SCT barrel layers (8), B = 2 T	paper Sec. III C (8 layers); IBL radius and length not stated in the paper, taken from arXiv:1803.00844
Barrel lengths	IBL $\|z\| < 33.0$ cm, pixel $\|z\| < 40.05$ cm, SCT $\|z\| < 74.9$ cm	Ref. [27]
Pixel hits	pixel centre, 50 × 400 μm (IBL 50 × 250 μm); neighbouring pixels merged	paper Sec. III A
SCT R-φ	34 μm (2 × 17 μm)	paper Sec. III A + Ref. [27]

Item	This work	Source
SCT z (signal)	**1.16 mm** and **no smearing**, both reported	see Sec. 1.4
Beam spot	45 mm Gaussian, one common shift per event; plane fit from the true vertex	paper Sec. III A, III C; per event vs per hit not stated, our choice
Hits	first crossing of each layer by each quirk, t < 25 ns; every crossing shown as a sensitivity	paper does not state how repeated crossings are treated; our choice
Plane finding	seeds, windows, factor-3 rule, Δw < 1 cm, smallest Δs; Δs = $\sqrt{\lambda_1}$, Δw = $\sqrt{\lambda_2}$ of Eq. 11	paper Sec. III C
Signal region	plane; 2 hits in 7 layers and ≥ 1 in the 8th; Δw < 1 cm; Δs < 0.01 cm	paper Sec. IV

ε_{trig} = N(pT(QQ̄) > 200 GeV) / N(after matching) ε_{fid} = N(both quirks cross all 8 layers inside the lengths, t < 25 ns) / N(triggered; colored: both charged) ε_{reco} = N(signal region) / N(fiducial) $\varepsilon = \varepsilon_{trig} \times \varepsilon_{fid} \times \varepsilon_{reco}$ (paper Eq. 12)

1.4 SCT z values used

Paper, Sec. III A

> "for the silicon microstrip tracker (SCT) we conservatively assume a resolution of twice the intrinsic accuracy quoted in [27]."

Ref. [27], ATL-PHYS-PUB-2009-002 (in arXiv:0901.0512, p. 18)

> "The intrinsic accuracies per module in the barrel are 17 μm (R−φ) and 580 μm (z)"

- **Signal.** The paper's text gives 1.16 mm (2 × 580 μm). Its Fig. 5 and its Λ column are matched only without SCT z smearing. Both settings are reported throughout.
- **Backgrounds.** The paper processes pile-up and conversions with the simplified detector of [34]. [34] takes its resolutions from the ATLAS technical design report but gives no SCT z value; we use 580 μm from [27], together with the energy-loss model of [34]. The backgrounds are the same for both signal settings.

2. Table I: signal efficiencies

2.1 Event counts (2000-event samples, first crossing)

Step	Colored, 1.16 mm	Colored, no smear	EW, 1.16 mm	EW, no smear
after matching	873	873	1045	1045
triggered	207	207	186	186
both charged	68	68	—	—
fiducial	22	22	62	62
plane found (Δw < 1 cm)	9	18	45	62
+ hit multiplicity	4	8	24	50
+ Δs < 0.01 cm (signal region)	1	8	10	50
signal region, every crossing as a hit (≥ 2 hits)	2	11	12	49

Quirks often cross a layer more than once within 25 ns. The italic row counts every crossing as a hit. 2 colored and 1 EW triggered events stopped with a numerical integration error; they are kept in the ε_{fid} denominator as not fiducial.

2.2 Efficiencies against the paper

Benchmark	Efficiency	k / n	ε [Wilson 95%]	Paper	p	Verdict
Colored 1.8 TeV, 2 keV	ε_{trig}	207 / 873	0.237 [0.210, 0.266]	0.24	0.87	**No significant inconsistency**
	ε_{fid}	22 / 68	0.324 [0.224, 0.442]	0.28	0.42	**No significant inconsistency**
	ε_{reco}, SCT z 1.16 mm	1 / 22	0.045 [0.008, 0.218]	0.35	0.001	**Strongly inconsistent**
	ε_{reco}, no SCT z smearing	8 / 22	0.364 [0.197, 0.570]	0.35	1.00	**No significant inconsistency**
	ε (Eq. 12), 1.16 mm		0.0035 [0.0004, 0.0156]	0.024*		
	ε (Eq. 12), no smear		0.0279 [0.0132, 0.0494]	0.024*		
EW 800 GeV, 4 keV	ε_{trig}	186 / 1045	0.178 [0.156, 0.202]	0.10	1.8×10^{-14}	**Strongly inconsistent**
	ε_{fid}	62 / 186	0.333 [0.270, 0.404]	0.28	0.12	**No significant inconsistency**
	ε_{reco}, SCT z 1.16 mm	10 / 62	0.161 [0.090, 0.272]	0.72	7.8×10^{-20}	**Strongly inconsistent**
	ε_{reco}, no SCT z smearing	50 / 62	0.806 [0.691, 0.886]	0.72	0.16	**No significant inconsistency**

Benchmark	Efficiency	k / n	ε [Wilson 95%]	Paper	p	Verdict
	ε (Eq. 12), 1.16 mm		0.0096 [0.0049, 0.0167]	0.020*		
	ε (Eq. 12), no smear		0.0478 [0.0357, 0.0614]	0.020*		

* derived: product of the paper's Table I entries (the paper does not quote ε). Interval of our ε: 95% range of the product of the three binomial factors (Jeffreys draws). Colored: the both-charged requirement (68 of 207 triggered events) is applied before ε_{fid}, so it is not part of Eq. 12.

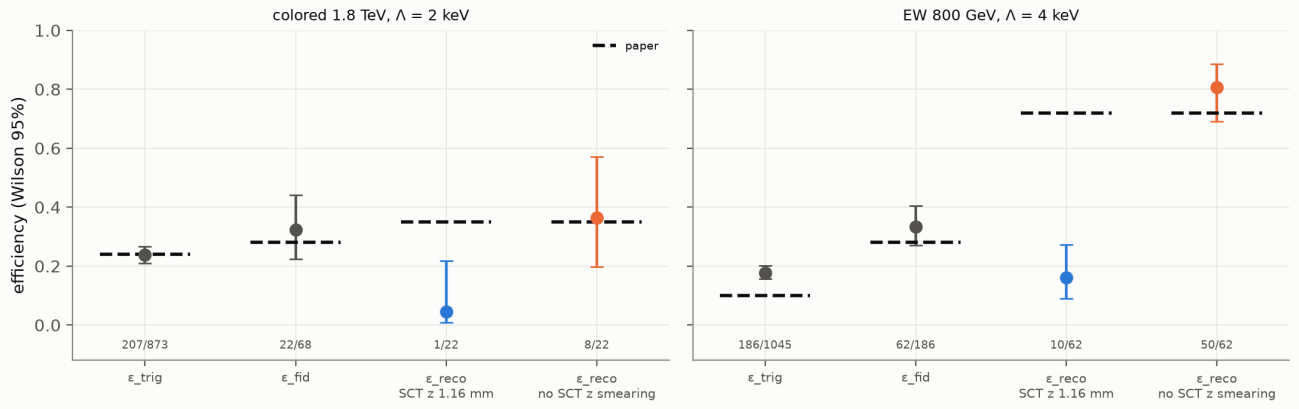


Table I efficiencies: points = this work with Wilson 95% intervals, dashes = paper. Numbers below the points: k/n.

What this shows: where a point's bar crosses the dashed line, our sample agrees with the paper. The colored trigger and fiducial and both fiducial values agree; the EW trigger is too high; ε_{reco} agrees for both samples only without SCT z smearing (orange) and is far too low with 1.16 mm (blue).

2.3 The Λ column of Table I

The paper also gives ε_{reco} at other string tensions Λ. The size of the quirk oscillation scales as $1/\Lambda^2$, so this column tests the reconstruction over a wide range of hit patterns. The fiducial count is recomputed at every Λ.

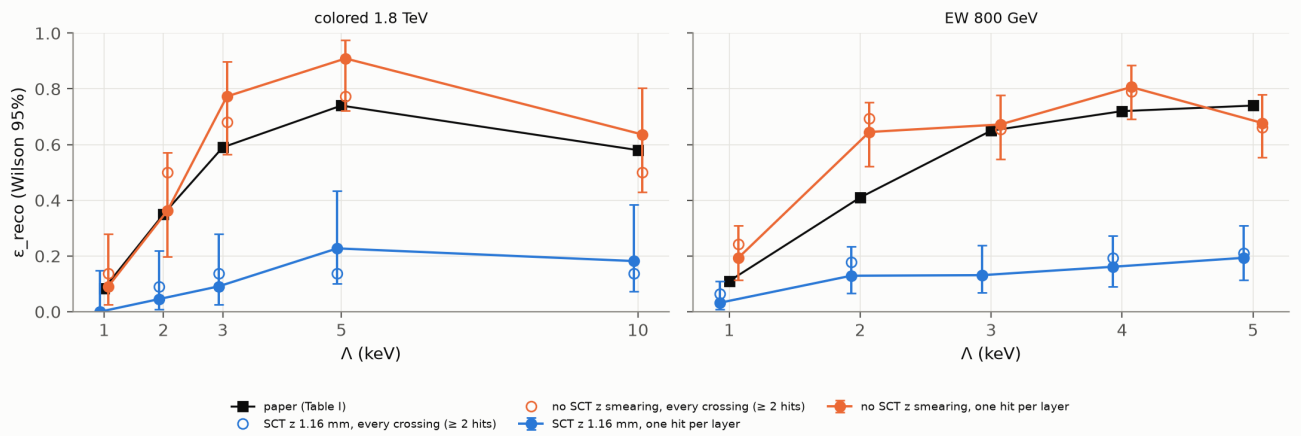
Benchmark	Λ (keV)	Paper	1.16 mm: k / n	ε [Wilson 95%]	p	no smear: k / n	ε [Wilson 95%]	p
Colored	1	0.083	0 / 22	0.00 [0.00, 0.15]	0.25	2 / 22	0.09 [0.03, 0.28]	0.70
	2	0.35	1 / 22	0.05 [0.01, 0.22]	0.001	8 / 22	0.36 [0.20, 0.57]	1.00
	3	0.59	2 / 22	0.09 [0.03, 0.28]	1.5×10^{-6}	17 / 22	0.77 [0.57, 0.90]	0.087
	5	0.74	5 / 22	0.23 [0.10, 0.43]	7.3×10^{-7}	20 / 22	0.91 [0.72, 0.97]	0.088
	10	0.58	4 / 22	0.18 [0.07, 0.39]	2.7×10^{-4}	14 / 22	0.64 [0.43, 0.80]	0.67
EW	1	0.11	2 / 62	0.03 [0.01, 0.11]	0.063	12 / 62	0.19 [0.11, 0.31]	0.042
	2	0.41	8 / 62	0.13 [0.07, 0.23]	1.9×10^{-6}	40 / 62	0.65 [0.52, 0.75]	2.4×10^{-4}
	3	0.65	8 / 61	0.13 [0.07, 0.24]	7.0×10^{-17}	41 / 61	0.67 [0.55, 0.78]	0.79
	4	0.72	10 / 62	0.16 [0.09, 0.27]	7.8×10^{-20}	50 / 62	0.81 [0.69, 0.89]	0.16
	5	0.74	12 / 62	0.19 [0.11, 0.31]	3.6×10^{-19}	42 / 62	0.68 [0.55, 0.78]	0.25

First crossing. Shading of the p column follows the verdict scale (green: no significant inconsistency).

Combined test of the whole Λ column

Setup (all 10 Λ points)	Colored χ^2 / 5	p	EW χ^2 / 5	p	Both χ^2 / 10	p	Verdict
SCT z 1.16 mm, first crossing	46.9	6.1×10^{-9}	141.2	9.8×10^{-29}	188.1	4.9×10^{-35}	**Strongly inconsistent**
no SCT z smearing, first crossing	4.4	0.50	23.8	2.4×10^{-4}	28.2	0.002	**Strongly inconsistent**
SCT z 1.16 mm, every crossing (>= 2)	64.2	1.7×10^{-12}	150.4	1.1×10^{-30}	214.5	1.5×10^{-40}	**Strongly inconsistent**
no SCT z smearing, every crossing (>= 2)	3.2	0.67	36.6	7.1×10^{-7}	39.8	1.8×10^{-5}	**Strongly inconsistent**

$\chi^2 = d^T C^{-1} d$ with d = (ours − paper) at every Λ; C uses binomial variances and the correlation between Λ points from an event-level bootstrap (the same events enter every Λ). Paper values taken as exact.



ε_{reco} vs Λ. Filled points: first crossing, Wilson 95%. Open points: every crossing as a hit (≥ 2 hits per layer). Blue: SCT z 1.16 mm; orange: no SCT z smearing; black: paper.

What this shows: the black line is the paper. Without SCT z smearing (orange) our points follow the paper's rise and fall with Λ; colored agrees (combined p = 0.50; 3 and 5 keV borderline), EW sits above the paper at 1–2 keV. With 1.16 mm (blue) the efficiency stays low at every Λ.

3. Backgrounds

Sample (detector: [34] model, SCT z 580 µm)	Events	Plane	+ hit mult.	Signal region k / n	Wilson 95%	Paper	p
converted γ + µ = 50 pile-up	9139	3399	419	0 / 9139	$[0, 4.2\times10^{-4}]$	"a few" (≈ 2.4×10^{-4})	0.28
pile-up only, µ = 50	5000	—	—	0 / 5000	$[0, 7.7\times10^{-4}]$	none	—

Pile-up plane rates (5000 events)	k / n	rate [Wilson 95%]	Paper	p (approx.)	Comparison
selected plane, ≥ 1 hit in ≥ 4 of 8 layers	33 / 5000	6.6×10^{-3} [4.7×10^{-3}, 9.3×10^{-3}]	$\sim10^{-3}$	9.8×10^{-17}	higher than the paper's approximate rate
selected plane, ≥ 1 hit in all 8 layers	1 / 5000	2.0×10^{-4} [3.5×10^{-5}, 1.1×10^{-3}]	6×10^{-5}	0.26	compatible with the paper's approximate rate

Converted photons at 300 fb⁻¹	σ (Z(νν)γ + j)	conversion probability	signal-region fraction, 95% UL	expected events, 95% UL
this work's cross section	0.211 pb	1%	3.3×10^{-4}	< 0.21
paper's cross section	~1 pb	1%	3.3×10^{-4}	< 0.98

Converted photons: Z(→νν)γ + jet, MET > 200 GeV, pT(γ) > 0.5 GeV; 10 000 generated, 9139 pass MET > 200 GeV after the shower. Conversion probability 1% from the paper's 1 pb → 10 fb. Measured sensitivities, same events: SCT z 1.16 mm without the [34] energy loss: 0/9139 conversions, 31/5000 pile-up 4-of-8 planes; background without SCT z smearing: 146/9139 conversions in the signal region.

4. Figures 2–5 of the paper

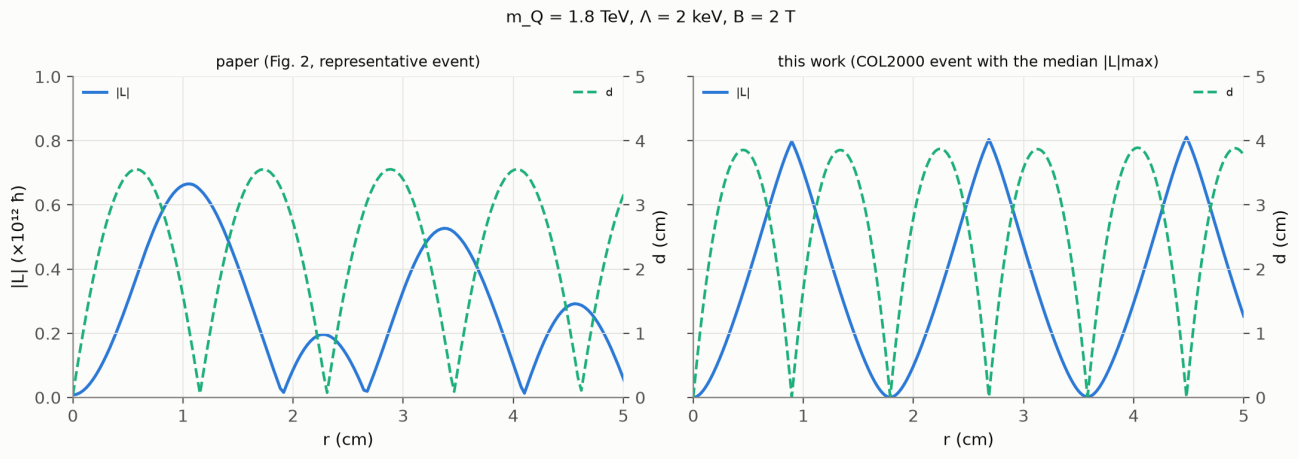
Paper curves and histograms are read from the vector figures in the paper's arXiv source, so the comparison uses the paper's exact values.

Figure 2: angular momentum and quirk separation

Fig. 2 quantity (colored 1.8 TeV, 2 keV)	Paper (one event)	This work (median-|L|max event)
maximum |L|	0.67×10^{12} ħ	0.81×10^{12} ħ
maximum separation d	3.56 cm	3.89 cm
spacing in r between minima of d	1.15 cm	0.90 cm
separation d at the closest approaches	not readable	limited by the time step: 37–186 µm at 300 steps per period, 0.5–1.7 µm at 30 000; still ≫ $1/m_Q = 1.1\times10^{-17}$ cm

Angular momentum at closest approach (all COL2000 fiducial events)	This work	Paper
events / closest approaches (r < 15 cm)	22 / 276	one event shown
closest approaches with |L| < 1% of the event's |L|max	132 / 276 = 0.48 [0.42, 0.54]	“whenever |L| = 0, the separation between the quirks is large”
events with at least one such approach	22 / 22	—
smallest |L| at a closest approach	4.5×10^{6} ħ (not converged: decreases with a finer time step)	—
|L| from the torque of Eq. 5, q(∫d dt) × E_{cm}, vs simulated |L|: largest relative difference, median over events	0.11	Eq. 5

Integration with 3000 steps per period (10× the production setting). At the large-|L| approaches |L| is stable under a finer step; at the small-|L| approaches it is not.

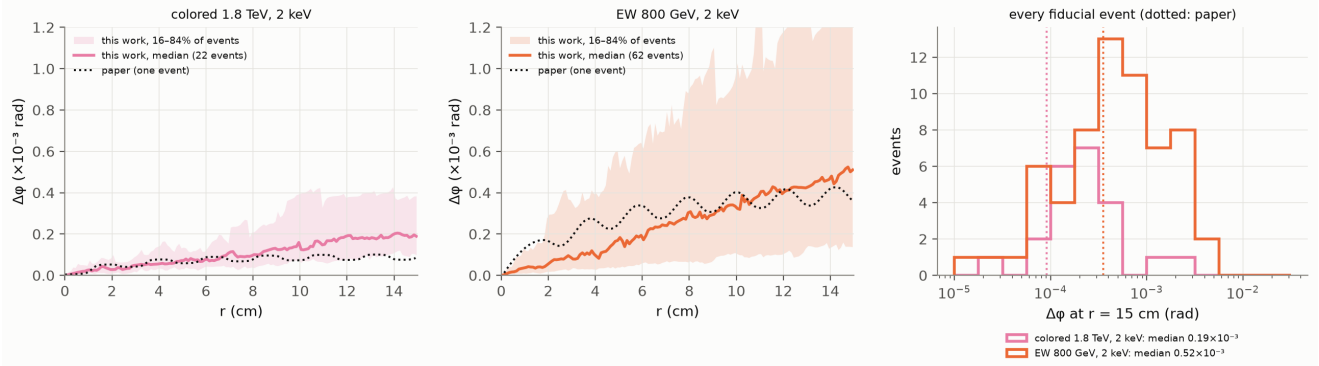


|L| (solid, left axis) and quirk separation d (dashed, right axis) in the exact centre-of-mass frame vs the radius r of the centre of mass.

What this shows: the two quirks fly apart and come back together again and again (dashed d). Our event and the paper's event have the same size of oscillation (3.5–3.9 cm) and the same scale of angular momentum (about 10^{12} ħ). In our events |L| falls close to zero at every second closest approach; in the paper's event it does not.

Figure 3: rotation of the quirk plane

Fig. 3 at r = 15 cm (Λ = 2 keV)	Events	Δφ median [16–84%]	Paper	our events below the paper: k / n [Wilson 95%]
colored 1.8 TeV	22	0.19 [0.10, 0.38]×10^{-3}	0.09×10^{-3}	2 / 22 = 0.09 [0.03, 0.28]
EW 800 GeV	62	0.52 [0.14, 1.70]×10^{-3}	0.36×10^{-3}	24 / 62 = 0.39 [0.28, 0.51]



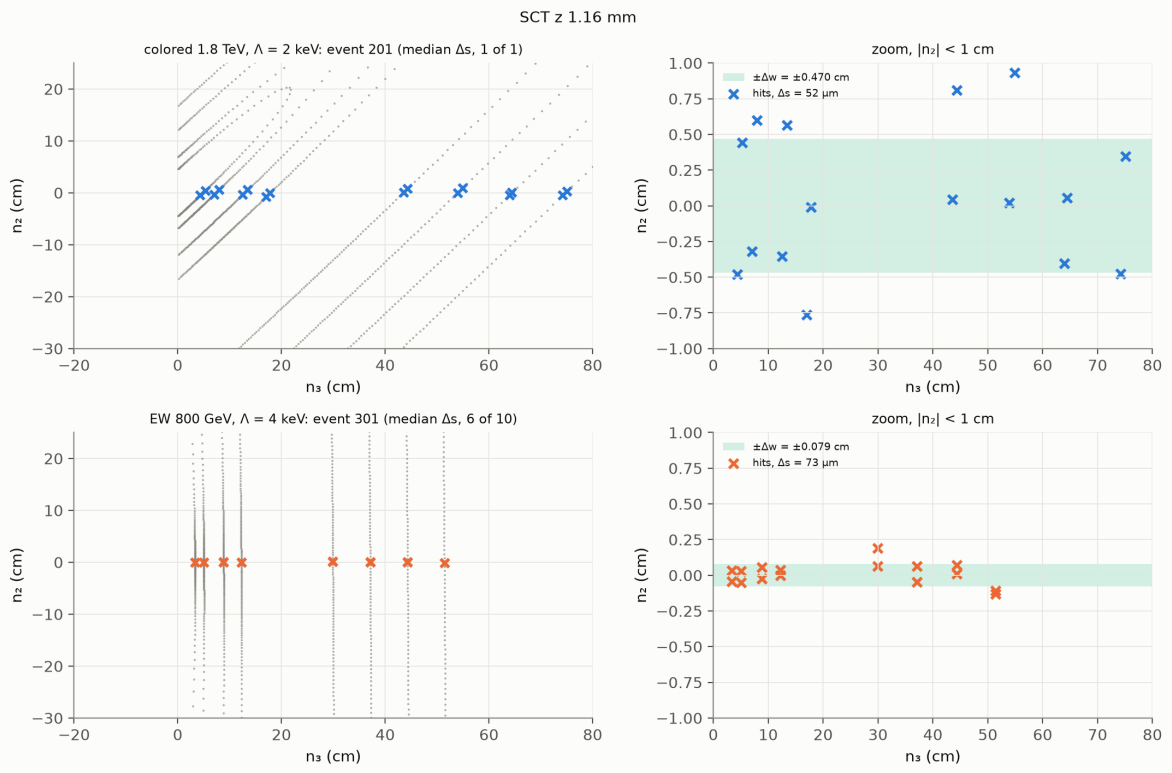
Δφ: angle by which the plane spanned by the two quirks' velocity vectors (the paper's definition) has turned, vs r. Line: median of all fiducial events; band: 16–84% of events; dotted: the paper's event. Right: Δφ at r = 15 cm for every event.

What this shows: the plane of the two quirks turns by less than about 10^{-3} rad over 15 cm, so the hits stay on one flat plane. The paper's single event lies inside our spread (at the low end for colored).

Figure 4: hits of one signal event in the plane's coordinates

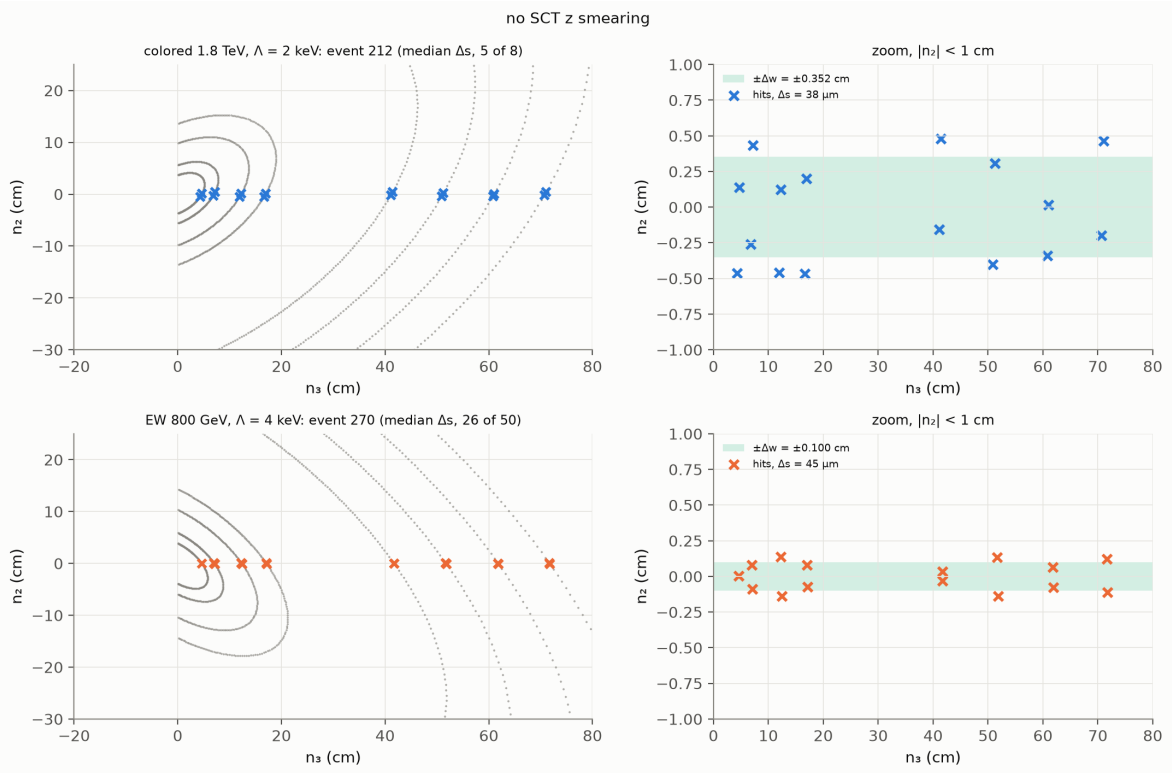
Fig. 4 example event (median Δs, signal region)	SCT z	Event	rank	Δs	Δw
Colored	1.16 mm	201	1 of 1	52 μm	0.470 cm
EW	1.16 mm	301	6 of 10	73 μm	0.079 cm
Colored	none	212	5 of 8	38 μm	0.352 cm
EW	none	270	26 of 50	45 μm	0.100 cm

Fig. 4 example event (median Δs, signal region)	SCT z	Event	rank	Δs	Δw
paper example (colored)	—	—	—	83 μm	0.074 cm



SCT z 1.16 mm. The event is chosen by a fixed rule: the signal-region event with the median Δs.

What this shows: all hits lie on a narrow strip (right, zoom). Δs is the thickness of the plane and Δw the width of the strip (green band). With 1.16 mm only one colored event passes.



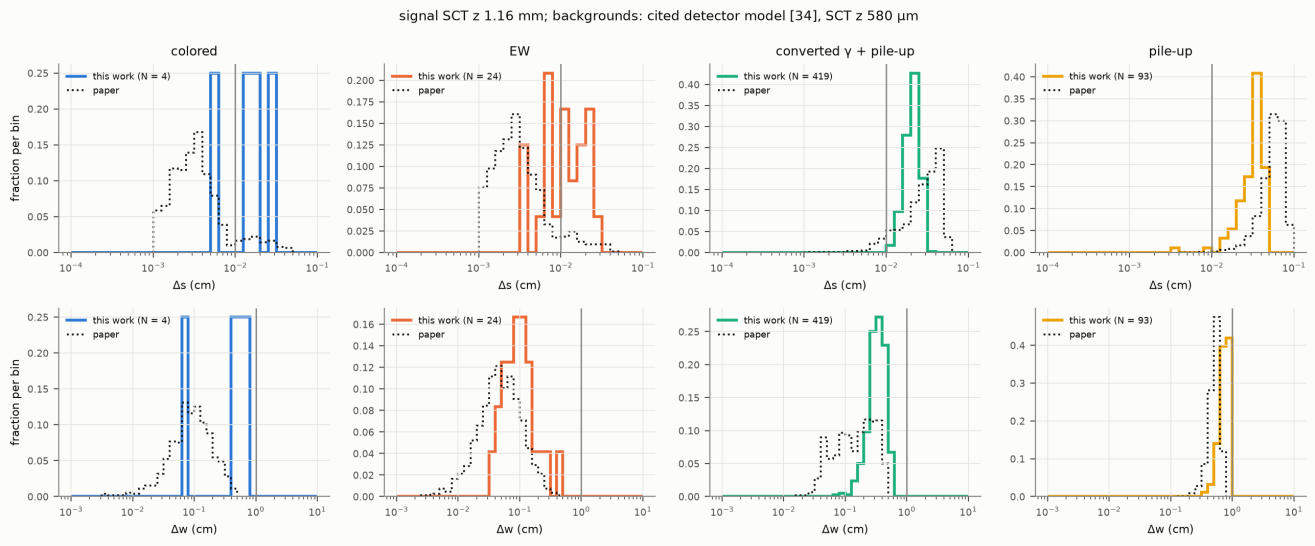
No SCT z smearing, same rule.

What this shows: without SCT z smearing the hits sit closer to the plane (smaller Δs), so many more events pass; the strip width is set by the quirk oscillation.

Figure 5: Δs and Δw distributions

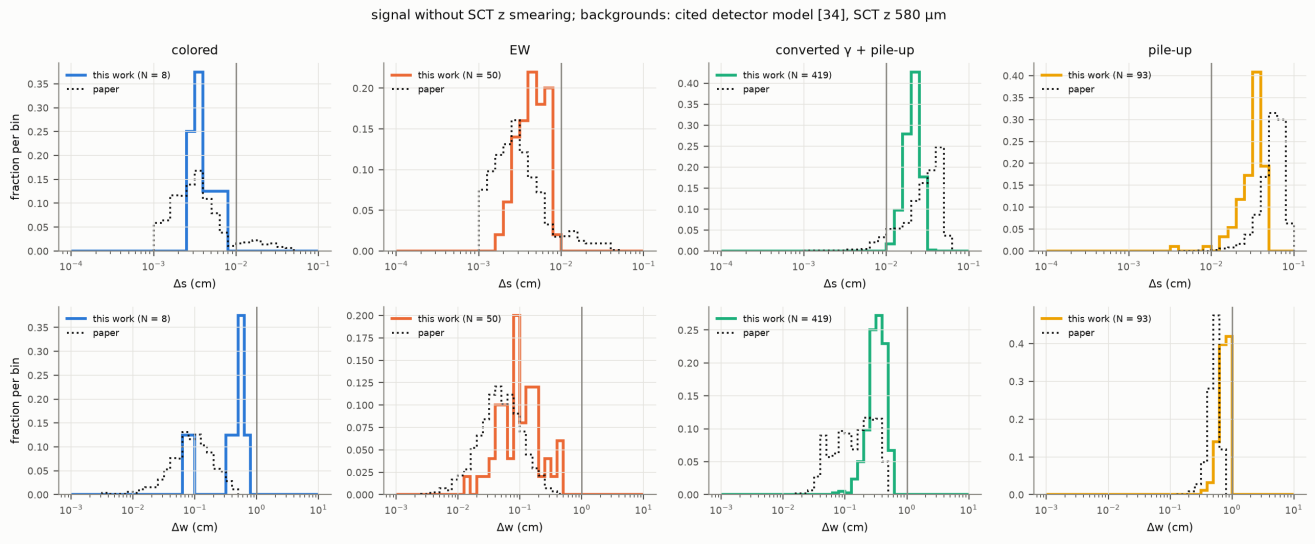
Population	SCT z (signal)	N	Δs median	paper	Δw median	paper	Δs < 0.01 cm: k / n	Wilson 95%	paper	p
colored	1.16 mm	4	160 µm	32 µm	0.550 cm	0.092 cm	1 / 4	0.25 [0.05, 0.70]	0.899	0.004
colored	none	8	38 µm	32 µm	0.482 cm	0.092 cm	8 / 8	1.00 [0.68, 1.00]	0.899	1.00
EW	1.16 mm	24	110 µm	28 µm	0.092 cm	0.048 cm	10 / 24	0.42 [0.24, 0.61]	0.916	7.2×10^{-10}
EW	none	50	44 µm	28 µm	0.097 cm	0.048 cm	50 / 50	1.00 [0.93, 1.00]	0.916	0.020
converted γ + pile-up	580 µm, [34]	419	212 µm	306 µm	0.340 cm	0.156 cm	0 / 419	0.00 [0.00, 0.01]	0.069	1.4×10^{-13}
pile-up	580 µm, [34]	93	340 µm	571 µm	0.751 cm	0.525 cm	2 / 93	0.02 [0.01, 0.08]	0.002	0.010

Signal: COL2000 / EW2000 events after the plane and hit cuts, before the Δs cut. Paper medians and fractions from its vector histograms. The p column tests the fraction below the Δs cut against the paper's fraction (paper taken as exact). Colored Δw ratio to the paper: 5× here; with the 1000- and 2000-event samples together, 3.8–4.1×.



Signal with SCT z 1.16 mm; backgrounds with the [34] model and SCT z 580 µm. Unit area; dotted: paper. Vertical lines: signal-region cuts.

What this shows: with 1.16 mm the signal Δs (top, blue and orange) lies at about 100–200 µm, mostly to the right of the cut at 0.01 cm, while the paper's signal lies at about 30 µm.



Signal without SCT z smearing; same backgrounds.

What this shows: without SCT z smearing the signal Δs sits at 40 µm, close to the paper, and all signal is left of the cut; almost all background stays to the right. The signal Δw (bottom) is wider than the paper's.

5. Remaining differences

Quantity	This work	Paper
EW ε_{trig}	186/1045 = 0.18	0.10
EW ε_{reco} at Λ = 2 keV (no SCT z smearing)	40/62 = 0.65	0.41
ε_{reco} with SCT z 1.16 mm	colored 0.05, EW 0.16	0.35, 0.72
signal Δw median (Fig. 5)	EW 0.097 cm (2×), colored 0.482 cm (5×)	EW 0.048, colored 0.092 cm
pile-up rate, plane with ≥ 1 hit in ≥ 4 of 8 layers	6.6×10^{-3}	$\sim10^{-3}$

The paper does not give the information needed to trace these differences.

Settings the paper does not state

- matching scale and matching settings; PDF and renormalisation/factorisation scales
- which SCT coordinate the doubled resolution applies to
- how repeated crossings of a layer by the same quirk are treated, and whether “2 hits” means exactly 2 or at least 2
- the plotting convention of Δw in Fig. 5
- the minimum-bias process and tune used for pile-up, and the per-event or per-hit beam-spot smearing

Appendix A: what each setting rests on

Setting	Value used	Status
Benchmarks, trigger, signal region, plane algorithm	as in the paper	stated in the paper
Pixel size, merging, beam-spot width, B field, 25 ns	as in the paper	stated in the paper
SCT R-φ	34 µm = 2 × 17 µm	stated in the paper + Ref. [27]
SCT z (signal)	1.16 mm and none	1.16 mm: paper text + Ref. [27]; none: our choice, matches Fig. 5
SCT z (background)	580 µm	our choice: [34] cites the TDR, value from Ref. [27]
Background energy loss	bremsstrahlung + ionisation, no multiple scattering	Ref. [34], App. A
Barrel lengths	Ref. [27] values	from Ref. [27]
IBL radius and length	3.3 cm, $\lvert z\rvert$ < 33.0 cm	not stated; arXiv:1803.00844
Repeated layer crossings	first crossing (every crossing as sensitivity)	not stated; our choice
Beam-spot shift	one common shift per event, fit from the true vertex	not stated; our choice
Generator settings (matching, PDF, scale, tune)	Sec. 1.1	not stated; our choice
Minimum-bias sample	Pythia 8 SoftQCD:nonDiffractive, Monash, 20 000 events	paper: Pythia8, 125 000 events; process and tune not stated
Conversion probability	1%	derived from the paper's 1 pb → 10 fb

Statistics

Wilson 95% interval for k of n (ε = k/n, z = 1.96): $[\varepsilon + z^2/2n \pm z\sqrt{(\varepsilon(1-\varepsilon)/n + z^2/4n^2)}] / (1 + z^2/n)$. p: exact two-sided binomial probability of a count as unlikely as ours if the paper's value were exact. 95% upper limit for 0 of n: $1 - 0.05^{1/n} \approx 3/n$.

References

[paper] S. Knapen, H. K. Lou, M. Papucci, J. Setford, Phys. Rev. D 96, 115015 (2017), arXiv:1708.02243.
[27] ATLAS, “The Expected Performance of the ATLAS Inner Detector”, ATL-PHYS-PUB-2009-002, in arXiv:0901.0512.
[34] S. Knapen, S. Pagan Griso, M. Papucci, D. J. Robinson, “Triggering soft bombs at the LHC”, arXiv:1612.00850, App. A.
MadGraph5_aMC@NLO: J. Alwall et al., JHEP 07 (2014) 079. Pythia 8: C. Bierlich et al., SciPost Phys. Codeb. 8 (2022). IBL: ATLAS, arXiv:1803.00844.